EEG-Based Mental Stress Detection and Biofeedback System

Final Year Project Proposal (FYP-ECE-26-042) — Simple-Language Version
Department of Electrical & Computer Engineering, Academic Year 2026–2027

1. Summary

This project builds a wearable headset that reads brain signals (EEG) to detect mental stress in real time. The headset has 8 sensors on the head, placed at Fp1, Fp2, F3, F4, C3, C4, P3 and P4 (a standard sensor layout called the 10-20 system), plus 2 ear clips. A chip called the ADS1299 turns the tiny brain signals into digital data 250 times per second.

A small computer cleans the data, removes noise from eye blinks and face muscles, and picks out useful patterns. A machine learning model then sorts the person's state into three levels: **Relaxed**, **Moderate (healthy) stress**, or **High distress**. The target is more than 85% accuracy. If high distress lasts, an app starts a guided breathing exercise to help the person calm down.

Total cost is about **$1,650 (Rs. 462,000)**, with a cheaper option of about **$520 (Rs. 145,600)**.

Main things the team will build

- **Headset:** a comfortable 8-sensor dry-electrode EEG headset for the front and top of the head.
- **Noise removal:** real-time filtering of eye-blink and facial-muscle noise without damaging the brain signals.
- **Feature extraction:** measures such as band power, Frontal Alpha Asymmetry (FAA) and stress ratios.
- **Machine learning:** a small, fast model with delay under 250 ms and accuracy above 85%.
- **Biofeedback app:** a dashboard with a live stress meter and automatic breathing guide.

2. The Problem

Stress is a major cause of long-term health problems, burnout and mistakes. Long-term stress can affect the brain areas used for decision-making and emotion, and can lead to high blood pressure, heart disease, weaker immunity and depression. In high-risk jobs such as aviation, surgery, the military and transport, unnoticed stress can cause slow decisions and serious errors. Stress also costs economies hundreds of billions of dollars each year through missed work, lower productivity and medical bills.

Why current methods fall short

Method	What it measures	Main problems	Delay
Questionnaires (PSS, DASS-21, STAI)	Self-reported answers	Depends on memory and honesty; cannot track stress live; interrupts work	Hours to weeks
Cortisol tests	Stress hormone in saliva or blood	Needs body fluid and a lab; costly; hormone peaks late	15-30 minutes
Smartwatches (heart rate, skin sweat)	Indirect body signs	Affected by exercise, heat, humidity, caffeine; cannot tell exercise from stress	1-5 minutes

Method	What it measures	Main problems	Delay
EEG (this project)	Brain signals directly	Direct and fast; not confused by exercise once filtered	Under 250 milliseconds

Why EEG works

The front part of the brain handles threat detection, judgment and emotion control. EEG records its electrical activity with very fast timing. Research links stress to these patterns:

- **Alpha waves (8-13 Hz) go down.** Alpha is a relaxed-state wave. Mental effort and anxiety reduce it.
- **Beta waves (13-30 Hz) go up.** Beta is a fast, alert wave. Stress raises it, especially at the forehead (Fp1, Fp2).
- **Frontal Alpha Asymmetry (FAA).** Under stress, the right front of the brain tends to be more active than the left. This shows as less alpha at F4 (right) than at F3 (left).

3. The Proposed System (3 Layers)

Layer 1: The headset

- 8 dry Ag/AgCl sensors (Fp1, Fp2, F3, F4, C3, C4, P3, P4) and 2 ear clips (A1/A2) for reference and ground.
- Texas Instruments ADS1299 chip: 24-bit, very low noise, 250 samples per second.
- Microcontroller and wireless link (2.4 GHz RF / Bluetooth LE) send the data onward with delay under 15 ms.

Layer 2: The small computer (Raspberry Pi 5, 8 GB)

- **Cleaning:** a 50 Hz notch filter removes mains hum; a 0.5-45 Hz bandpass filter keeps the useful brain range.
- **Artifact removal:** FastICA removes eye blinks; wavelet denoising (sym4) removes facial-muscle noise.
- **Features:** the signal is cut into 2-second pieces (50% overlap). For each piece it calculates band power (theta, alpha, beta, gamma), FAA and differential entropy.
- **Machine learning:** a compressed model (SVM, LightGBM or 1D-CNN) gives the result in under 25 ms: Class 0 Relaxed, Class 1 Moderate stress, Class 2 High distress.

Layer 3: Dashboard and breathing help

- Live stress meter (0-100), signal quality per channel and workload curves, built with PyQt6 or Streamlit.
- Automatic breathing guide when stress stays high.
- Data is stored locally in an encrypted SQLite database, with no cloud, to protect privacy.

Why these sensor positions

- **Fp1, Fp2 (forehead):** emotion control and threat judgment. Flat dry sensors work here because there is no hair.

- **F3, F4 (front left and right):** used to calculate FAA. The left side relates to positive, approach feelings; the right side to withdrawal and stress.
- **C3, C4, P3, P4 (top and back):** track movement-related rhythms and mental workload, helping the model tell plain concentration from emotional distress.
- **A1, A2 (earlobes):** reference and ground, with low contact resistance.

How the breathing help works

If the stress level stays above the set threshold for 4 epochs in a row (8 seconds), the app starts a breathing pacer: 4 seconds breathing in and 6 seconds breathing out (about 6 breaths a minute, or 0.1 Hz). This slow pace activates the vagus nerve and helps the body relax before performance drops.

4. How the Project Works: 5 Stages

Stage	What happens	Result
1. Record data	25 people do stress tasks using PsychoPy: the Stroop colour-word test and timed mental maths (MIST), plus a rest recording with eyes open and closed. EEG is recorded at 250 Hz.	Labelled raw EEG data from 25 people
2. Clean signals	Remove mains hum, filter 0.5-45 Hz, remove eye blinks with ICA, remove muscle noise with wavelets.	Clean brain-signal data
3. Find features	Cut into 2-second pieces. Calculate band powers, FAA, stress ratio, differential entropy and Hjorth parameters.	32 numbers per piece
4. Train models	Test SVM, Random Forest and 1D-CNN. Check on unseen people (leave-one-subject-out) and 10-fold validation. Tune settings and shrink the model (INT8).	Small model (under 2 MB), over 85% accuracy
5. Run live	Put the model on the Raspberry Pi using ONNX Runtime. Show live results and trigger breathing help.	Working prototype, delay under 250 ms

The main formulas, in plain words

1. **Band power (Welch's method):** measures how strong each brainwave band is: theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), gamma (30-45 Hz).
2. **FAA = ln(alpha at F4) - ln(alpha at F3).** Alpha is low when a region is active. A negative FAA means the right front brain is more active, which is linked to anxiety and withdrawal.
3. **Stress ratio = beta / (alpha + theta).** A higher value means more mental arousal and effort compared with resting waves.
4. **Differential entropy:** a measure of signal complexity, often better than plain power at spotting changes in thinking and emotion state.

5. Hardware and Software

Hardware

Part	Model	What it does	Qty
Signal board and ADC	OpenBCI Cyton (ADS1299)	8 channels, 24-bit, low noise, PIC32 microcontroller	1

Part	Model	What it does	Qty
Headset frame	Ultracortex Mark IV (open-source, 3D printed)	Adjustable frame that holds sensors in 10-20 positions (head 54-58 cm)	1
Scalp electrodes	Dry Ag/AgCl comb and flat disc	8 sensors: combs for F3, F4, C3, C4, P3, P4 (through hair), discs for Fp1, Fp2; 10 units in total	10
Ear electrodes	Gold-plated ear clips	Reference and ground	2
Wireless link	OpenBCI USB RF dongle	Fast 2.4 GHz streaming, delay under 15 ms	1
Edge computer	Raspberry Pi 5 (8 GB)	Runs filtering, ICA, features and the ML model	1
Battery	3.7 V 1200 mAh LiPo + protection board	Battery-only power for electrical isolation (IEC 60601-1)	2

Software

Area	Tools	Purpose
Data collection	BrainFlow, Lab Streaming Layer	Live streaming, buffers, timestamps, clock sync
Signal cleaning	MNE-Python, SciPy, PyWavelets	Filters, FastICA, wavelet denoising
Features	NumPy, SciPy, BioSPPy, Numba	Band power, FAA, Hjorth, differential entropy
Machine learning	Scikit-learn, PyTorch, ONNX / TFLite	Train, validate and compress SVM, Random Forest, CNN-LSTM
Dashboard	PyQt6 / Streamlit, Plotly	Live stress gauge, EEG traces, trends, breathing trigger
Stress tests	PsychoPy 3	Stroop and maths tasks with precise event markers in the EEG

6. Budget

Prices come from official suppliers (OpenBCI, Mouser, Adafruit, Raspberry Pi). The rate used is 1 USD = 280 PKR.

Category	Item	Qty	Total (USD)	Total (PKR)
Hardware	OpenBCI Cyton board + dongle	1	$699	Rs. 195,720
Hardware	Ultracortex Mark IV kit and straps	1	$150	Rs. 42,000
Hardware	Dry electrodes + ear clips	1 set	$120	Rs. 33,600
Hardware	Raspberry Pi 5 (8 GB), cooler, 64 GB card	1	$95	Rs. 26,600
Hardware	LiPo batteries + protection	2	$40	Rs. 11,200
Prototyping	High-quality 3D printing of frame	1 job	$80	Rs. 22,400
Consumables	Shielded leads, connectors, soldering kit	1 lot	$45	Rs. 12,600
Consumables	Skin prep gel, alcohol wipes, sanitizer	1 lot	$35	Rs. 9,800
Cloud	GPU time for model training (3 months)	3	$75	Rs. 21,000

Category	Item	Qty	Total (USD)	Total (PKR)
Human trials	Payment for 25 participants ($6 each)	25	$150	Rs. 42,000
Contingency	Spares, import duty, shipping	1 lot	$161	Rs. 45,080
TOTAL			**$1,650**	**Rs. 462,000**

Low-cost option (about $520 / Rs. 145,600)

Because BrainFlow hides hardware differences, the OpenBCI board can be swapped for either a 4-channel Muse S / Muse 2 headset (about $250, Bluetooth LE) or a custom 4-channel ADS1299 board made at JLCPCB (about $80 for 5 assembled boards). Add an ESP32-S3 host (about $15). This cuts cost by more than 68% and still supports FAA using Fp1, Fp2, F3 and F4.

7. Timeline (36 Weeks)

Semester	Weeks	Work	Deliverable
1	1-6	Read research, finalize sensor positions, buy hardware	Approved specification and parts list
1	7-12	3D print frame, assemble, wire electrodes, flash firmware, check signal quality and impedance	Working headset with clean data stream
1	13-18	Build PsychoPy tests (Stroop, MIST), pilot on 5 people, mid-term defense	Mid-year report and live streaming demo
2	19-24	Collect data from 25 people; apply filters and ICA	Clean, labelled dataset
2	25-30	Extract features, train SVM / RF / 1D-CNN, 10-fold validation, INT8 compression	Validated model (over 85% accuracy)
2	31-36	Build dashboard, test breathing help, measure latency, write thesis, final defense	Full prototype, thesis and presentation

8. Risks and Solutions

- **Blinks and movement noise (worst at the forehead):** FastICA removes blink components; wavelet denoising removes muscle noise.
- **Brains differ between people:** each session starts with a 60-second rest recording (eyes open and closed), so the system uses changes from the person's own baseline instead of raw values.
- **Slow processing:** INT8 compression and multi-threaded buffers keep total delay under 250 ms.

9. Ethics, Safety and Privacy

- The university ethics board (IRB) must approve all human trials before they begin.
- Participants give informed consent and can withdraw at any time.
- The headset runs only on an isolated 3.7 V battery with no connection to mains power, following IEC 60601-1 safety standards.
- All brain data is anonymized, encrypted with AES-256 and processed on the device itself.

Approval signatures

Lead Investigator	Project Supervisor	Head of Department / ECE Chair
_____ Signature & ID Date: _____	_____ Faculty Advisor Date: _____	_____ Capstone Committee Date: _____